GenAI for Engineers — Learner Guide

IES | v4.0 | Lesson-plan aligned | PPT slides 1–186

This guide follows the supplied “Lesson Plan with Hands-On Labs (v2 — PUB Feedback)” module sequence, all 16 named activities and learning outcomes. Detailed procedures live here; the presentation uses concise visual explanations. The activity folders are short connected tasks inside the six module time boxes.

Learning outcomes

Outcome	Learner evidence
LO1	Explain FMEA, Six Sigma/DMAIC and SPC before applying them.
LO2	Automate engineering documents and analyses with Copilot in Word, Excel and PowerPoint.
LO3	Apply R.I.S.E. and E-MAP with iterative prompting.
LO4	Build and deploy a multi-step Copilot Agent.
LO5	Use Researcher and Analyst for cited research and data analysis.
LO6	Ground an agent on SharePoint with cited, traceable answers.
LO7	Apply the complete 7-Step AI Security Framework.

One-day course map

Time	Module	PPT	Primary activity
09:00–10:00	1 · The Copilot Ecosystem for Engineers	Slides 5–25	Choose the right Copilot surface; complete access check and AI-vs-engineer challenge.
10:00–11:15	2 · Prompting for Engineering Documents	Slides 26–55	Apply R.I.S.E. and E-MAP, constraints, multi-persona review and iterative prompting.
11:30–13:00	3 · Engineering Methodology Primer & Document Automation	Slides 56–94	Explain and apply FMEA → DMAIC → SPC with three small Copilot in Excel labs.
14:00–15:15	4 · Copilot Agents & Microsoft 365 Reasoning Agents	Slides 95–124	Build and test a multi-step FMEA Agent; use Researcher and Analyst.
15:30–16:15	5 · SharePoint Integration with Copilot	Slides 125–143	Create/inspect a private knowledge base, ground an agent and verify citations.
16:15–16:45	6 · AI Security & Engineering AI	Slides 144–156	Apply all seven security controls

	Toolkit		and make a release decision.
16:45–17:00	Closing and Q&A	Slides 157–162	Toolkit and next action

Lesson Plan activity map

Module	Named activities
1	Lab 1.1 Access check; Lab 1.2 AI versus Engineer
2	Lab 2.1 Technical specification; Lab 2.2 Engineering report; Lab 2.3 Risk assessment and method statement
3	Lab 3.1 FMEA; Lab 3.2 DMAIC with SPC/Cp-Cpk; Lab 3.3 Technical specification pack
4	Lab 4.1 FMEA Assistant; Lab 4.2 Report Writer; Lab 4.3 Minutes workflow; Lab 4.4 Researcher and Analyst
5	Lab 5.1 SharePoint knowledge agent; Lab 5.2 Grounded agent chain
6	Lab 6.1 Seven-step security; Lab 6.2 AI Engineering Toolkit

Module 1 — The Copilot Ecosystem for Engineers



Learning outcome: choose the appropriate Microsoft 365 Copilot surface for a required engineering output and state the engineer's verification responsibility.

AI, GenAI and Microsoft 365 Copilot

Artificial intelligence performs tasks associated with human judgement. Generative AI creates a new draft from instructions and context. Microsoft 365 Copilot provides generative-AI assistance inside Microsoft 365 applications and through Copilot Chat and agents. A fluent answer is still a draft until its source, calculation and meaning are checked.

Surface	Use it when the output is…	Engineer verifies
Copilot Chat	A question, brief, plan or comparison	Evidence, assumptions and scope
Word	A specification, report, SOP or method statement	Requirements, clauses, numbers and approval
Excel	A table, formula, summary or chart	Rows, formulas, denominators and interpretation
PowerPoint	A decision story for an audience	Every value and conclusion
Agent	A repeatable multi-step output	Instructions, tests, permissions and release
SharePoint agent	A cited answer from internal documents	Source passage, revision and access

Activity 1.1 — onboarding and access check

- Sign in at https://m365.cloud.microsoft/chat with the approved work or school account.
- Open Word and Excel and confirm the Copilot control is available.
- Open Agents in Microsoft 365 Copilot.
- If a surface is unavailable after two minutes, record the limitation and use the trainer's worked example or paired screen.

Activity 1.2 — AI versus engineer challenge

Input: “Later pump tests are low. The strainer was replaced. A verification check also occurred. Pilot results improved.”

- One group manually separates facts, assumptions and questions.
- One group asks Copilot Chat for the same structure.
- Compare speed, completeness, unsupported claims and unanswered questions.
- Accept only the statements that can be traced to the input.

Your two class agents

Two Microsoft 365 Copilot agents are shared with the class for the whole day. Both explain and teach; neither approves an engineering decision. Open them from Copilot Chat, then Agents.

IES Engineering Methods Tutor — ask it anything about FMEA, DMAIC or SPC in your own words. Useful questions: why is a high Detection score bad news, what is the difference between percentage points and per cent, and why must control limits be frozen before monitoring.

IES AI Security Advisor — ask it about safe and accountable AI use: PDPA and personal data in prompts, data leakage routes, data ownership, the data lifecycle, sensitivity labels, and who remains accountable when an agent does the work.

Use them when you are stuck rather than guessing. Neither agent will do your lab for you; both will give you the method and a worked example and then ask you to try it.

Module 2 — Prompting for Engineering Documents



Learning outcome: build and iteratively improve a reviewable engineering prompt using R.I.S.E., E-MAP, constraints, personas and acceptance tests.

Why prompt engineering improves productivity

Prompt engineering reduces clarification cycles, makes repeat work more consistent, exposes assumptions, and asks for evidence in a format that is faster to review. It cannot compensate for missing or poor-quality source data. The practical goal is a checked result with less rework.

Principle	Question to answer	Example
Purpose	What decision or deliverable is needed?	Create a four-row FMEA review.
Evidence	Which exact source may be used?	Use only FMEAInput.
Context	What system, units and lifecycle apply?	CW-401 delivery test in L/h.

Steps	What work must happen and in what order?	Audit → calculate → chart → check.
Constraints	What must be preserved or avoided?	Do not alter ratings or claim cause.
Acceptance	How will the output be checked?	FM01 RPN must equal 140.

R.I.S.E. framework

Element	Meaning	Engineering use
R — Role	Perspective and vocabulary	“Act as an FMEA facilitator,” not an approving engineer.
I — Input	Evidence, rows, revision and scope	Use only the named table and preserve source rows.
S — Steps	Ordered calculations and operations	Audit, calculate, compare, present, check.
E — Expectations	Format, limits and acceptance test	Show formulas; state unknowns; return a fixed table.

E-MAP framework

Element	Meaning	Example
E — Engineering	System, function, lifecycle and decision	Chemical dosing skid; standardised delivery test.
M — Metrics	Formula, units, target and denominator	Deviation rate = deviations ÷ opportunities.
A — Assumptions	Known, unknown and provisional statements	Comparable test conditions are assumed and must be checked.
P — Parameters	Limits, revision, operating conditions	Eight-row baseline; TR-401 revision 1.

Constraints and acceptance tests

A useful constraint describes observable behaviour. “Be accurate” is difficult to test. “Preserve four supplied rows and ratings, show the RPN formula, flag Severity 9 or higher, and do not approve the design” is specific and reviewable.

Constraint type	Useful wording
Data	Preserve units, time order, missing values and source rows.
Evidence	Use only named documents or tables; cite each factual claim.
Calculation	Show formulas, source ranges and parameters.
Decision	Do not infer root cause, compliance, safety or approval.
Output	Return Observation | Evidence | Limitation | Next check.

Multiple personas and debate

Personas are useful when each represents a real responsibility. Ask each persona to review the same evidence, challenge another persona, and preserve disagreement rather than forcing consensus.

Run a two-round review. Reliability, Operations and Safety each state one claim with evidence. In round two, each challenges one other claim. Finish with agreed facts, unresolved disagreements and the smallest test that could resolve them. Do not force consensus.

Persona	Review question
Reliability	Which failure mechanism is unsupported?
Operations	Which operating or test condition may have changed?
Safety	What must be checked before changing equipment or controls?
Quality	Which criterion, record or calculation is missing?

Iterative prompting procedure

- Run the initial RISE/E-MAP prompt.
- Compare the output with one source passage, row and calculation.
- Select one material error or omission.
- Add one precise instruction that addresses that gap.
- Rerun and compare only the affected section.
- Record the correction and remaining limitation.

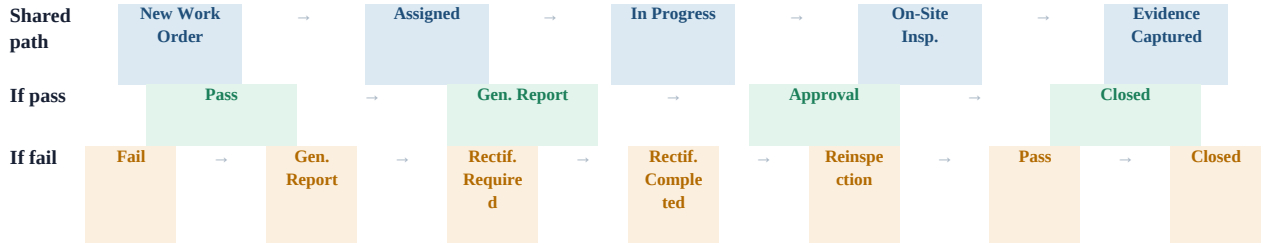
Module 2 document demonstrations

Output	Copilot surface	Prompt focus	Required check
Technical specification	Word	Requirements, tolerances, tests and clause checklist	Every “shall” traces to an approved requirement
Inspection report	Word	Observations, findings, recommendations and executive summary	A finding has a stated criterion
Risk assessment + method statement	Word	E-MAP context, hazards, controls and safe sequence	Approved matrix and competent-person sign-off
Management summary	PowerPoint	One headline, chart, evidence,	No value is introduced beyond

		limitation and decision	checked sources

Lab 2.1 — work order intake and assignment (30 minutes)

Open courseware/Labs/Lab-2.1-Technical-Specification/Starter.xlsx. Two fictional roadside drainage work orders are already formatted as the named Excel table WorkOrders, with the amber columns left for you to fill. Labs 2.1, 2.2 and 2.3 follow one work order journey: New Work Order → Assigned → In Progress → On-Site Inspection → Evidence Captured → Pass / Fail. This lab covers that shared path up to the initial decision. WO-021 will take the pass branch and WO-022 the fail branch, both in Lab 2.2. The inspection criterion for both assets is silt depth of at most 50 mm.



Step 1 — open the starter workbook. Choose Enable Editing if Excel shows the Protected View bar. The workbook opens on a Start_Here sheet naming the first action and explaining the rest: Work_Orders holds the supplied orders and the amber columns you fill, Status_History is the empty table for your status rows, Workflow_Guide restates the journey, and Copilot_Prompt carries the same prompts as the handout. The data is already a formatted Excel table, so nothing needs importing. Save a copy as Work-Order-Register.xlsx and work in that copy.

Step 2 — select the named table. Click any cell inside the work order data and confirm the Name Box, at the left of the formula bar, shows WorkOrders. Copilot in Excel works against the table you have selected, so an unselected or wrongly named table is the most common reason a prompt returns nothing useful. The workbook opens on a Start_Here sheet that names this first action and explains every other sheet.

Step 3 — run the Excel prompt. With the table selected, open Copilot from the Home ribbon and paste the Excel prompt printed below, or copy it from the Copilot_Prompt sheet without leaving Excel. It asks for a Status_History table with the six pre-decision rows per order and fills Initial_Outcome from the 50 mm rule. Copilot proposes; you confirm every row before accepting it.

Step 4 — build the status history properly. Create the six rows for each order in order: New Work Order, Assigned, In Progress, On-Site Inspection, Evidence Captured, then Pass or Fail. Use one row per status change. Never overwrite history with only the latest status — a reviewer has to be able to see the sequence. Leave unsupplied timestamps blank and write NOT SUPPLIED in Review_Note.

Step 5 — check the decision by hand. WO-021 reads 32 mm against the 50 mm minimum, so it passes with 18 mm to spare. WO-022 reads 74 mm, so it fails, 24 mm over the limit. Do this arithmetic yourself before trusting any formula. A reading is not a cause analysis: WO-022 failing tells you nothing yet about why.

Step 6 — draft the inspection plan in Word. Open a new Word document, open Copilot, attach the register, and paste the Word prompt below. Save as Inspection-Plan.docx labelled DRAFT FOR ENGINEER REVIEW. Then reconcile it against Excel: the same work order IDs, assets, locations, workers, criterion and evidence IDs must appear in both. The supplied mock-data.csv is only needed for the optional import practice in Appendix A of the lab handout.

Independent checks: WO-021 is 32 ≤ 50, Pass, inside by 18 mm; WO-022 is 74 > 50, Fail, over by 24 mm; each order carries six ordered pre-decision status rows; every missing permit, calibration detail or sign-off is marked NOT SUPPLIED.

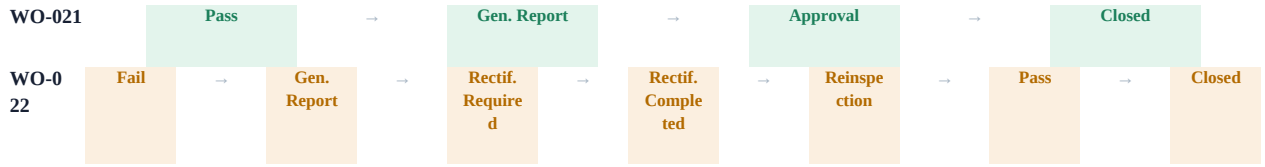
Boundary: this is a drafting aid, not an approval. The plan must not invent real approvals, photographs, calibration certificates or safety permits, and must not claim that site access was verified.

Use only the selected WorkOrders table on the Work_Orders sheet. Create a second table called Status_History with the columns Work_Order, Sequence, Status, Time_SGT, Actor, Source, Review_Note. For each of the two work orders add these six rows in this

order: New Work Order, Assigned, In Progress, On-Site Inspection, Evidence Captured, then the initial Pass or Fail. Use only the supplied timestamps; leave any time that was not supplied blank and write NOT SUPPLIED in Review_Note. In the WorkOrders table, fill Initial_Outcome as Pass when Initial_Reading is greater than or equal to 50 mm, otherwise Fail, and show the comparison you used. Preserve every ID, name, asset, location, unit and criterion source exactly. Do not invent evidence, approvals, permits, photographs or completed field work. End with DRAFT FOR ENGINEER REVIEW.

Lab 2.2 — inspection reports and rectification (30 minutes)

Open courseware/Labs/Lab-2.2-Engineering-Report/Starter.xlsx. Three inspection rows are formatted as the named table Inspections, and the two fictional WO-022 rectification events as RectificationLog. Both orders continue from the Pass / Fail decision you reached in Lab 2.1. WO-021 takes the pass branch: Pass → Generate Report → Approval → Closed. WO-022 takes the fail branch: Fail → Generate Report → Rectification Required → Rectification Completed → Reinspection → Pass → Closed.



Step 1 — open the workbook and orient yourself. The workbook opens on a Start_Here sheet. The others are Inspections (the three supplied rows plus the amber columns you fill), Rectification_Log, Branch_Guide, Check_Your_Work and Copilot_Prompt. The data is already a formatted Excel table, so nothing needs importing. Either continue here or copy these sheets into the Work-Order-Register.xlsx from Lab 2.1 so one file holds the whole journey, keeping the Lab 2.1 status history intact.

Step 2 — fill the inspection table with Copilot. Select the Inspections table, open Copilot, and paste the Excel prompt below. It computes Margin_mm and classifies each row against the 50 mm minimum. Keep the original readings visible: the reinspection is a new row, never an edit to the failed one.

Step 3 — verify the arithmetic by hand. WO-021 initial is 50 − 32 = +18, Pass. WO-022 initial is 50 − 74 = −24, Fail. WO-022 reinspection is 50 − 28 = +22, Pass. Confirm all three before accepting any Copilot output. The Check_Your_Work sheet lists the same figures.

Step 4 — generate RPT-021, the pass branch. Open a new Word document, open Copilot, attach the workbook and paste the Word prompt below. Include the order and asset, criterion and source, timestamp, reading, evidence reference, factual observation, interpretation, limitations, action and a blank reviewer decision. Save as RPT-021.docx. Record Generate Report as a status row, and record Approval only after a person checks the draft — the mock approval entry in the event log is classroom evidence, not a real approval.

Step 5 — generate RPT-022, the initial failure. Produce the WO-022 failure report the same way, preserving the 74 mm reading, EV-022-1 and the 24 mm exceedance exactly. Do not infer a cause from the reading: a silt depth above the limit is an observation, and the cause remains unproven. Save as RPT-022.docx, then record Rectification Required as a separate status row after the report exists.

Step 6 — add the reinspection addendum. The rectification log records a fictional desilting. Append a reinspection section carrying 28 mm and EV-022-2, and version the document as RPT-022-v2. Keep the original failed reading and the first report visible: a pass on reinspection does not erase the initial failure and does not prove the root cause was found. Record Rectification Completed, Reinspection, Pass, then Closed in that order.

Step 7 — reconcile and save. Reconcile every report ID and evidence ID against Excel. Add a reviewer note wherever evidence images, calibration records or real approvals are not supplied. Save the workbook and both reports as classroom drafts for Lab 2.3.

Independent checks: the three variances are +18, −24 and +22 mm; the pass branch shows report generation and approval before closeout; the fail branch shows a failure report before rectification, then a new reading before Pass and Closed; the first failed reading is still visible in the versioned document.

Boundary: no AI output is a real engineering approval. Do not assert an unproved root cause, and leave every real sign-off blank.

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Use only the selected Inspections table on the Inspections sheet. Preserve every Work_Order, Inspection_Round, timestamp, worker, asset, reading, unit, minimum, evidence ID and source status exactly as supplied. Add Margin_mm = Silt_Depth_mm − Maximum_mm for each of the three rows and show the arithmetic. Set Outcome to Pass when Silt_Depth_mm is greater than or equal to Maximum_mm, otherwise Fail. Keep all three rows visible, including WO-022's failed first inspection and its passing reinspection, and do not overwrite the first reading with the second. Mark missing real-world evidence as NOT SUPPLIED. Do not invent a root cause. End with DRAFT FOR ENGINEER REVIEW.
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Lab 2.3 — email the report to the manager (25 minutes)

Open your Work-Order-Register.xlsx from Labs 2.1 and 2.2, and this lab's Starter.xlsx for the approval log and recipient list. A report on your laptop closes nothing: Approval is a real gate between Generate Report and Closed, and somebody with authority has to pass through it. Here you request it, in Outlook, with Copilot drafting and you checking.

Step 1 — bring the approval sheets into your register. Copy Approval_Log, Recipients and Before_You_Send from Starter.xlsx into your register with Move or Copy, then close the starter without saving. Check you still have RPT-021.docx and RPT-022-v2.docx from Lab 2.2; you attach them here.

Step 2 — read the recipient list. The maintenance manager is asked to decide. The drainage supervisor and the records inbox are copied for information only. Say so in the email, so nobody assumes somebody else is approving.

Step 3 — draft the WO-021 email. Start a new mail, attach RPT-021.docx, open Draft with Copilot and paste the first prompt. It states DR-17 on Jalan Besar, 32 mm against the 50 mm maximum, the first-visit pass and evidence EV-021-1, and asks the manager to approve or return.

Step 4 — draft the WO-022 email. A separate new mail, attaching RPT-022-v2.docx, not the original. The prompt requires the 74 mm failure, the desilting, and the 28 mm reinspection pass, in that order, with both evidence IDs.

Step 5 — check both drafts before sending. The likely failure is that the WO-022 email reports only the passing reinspection. Asked to summarise, a model drifts toward the tidy ending, and a manager who cannot see the failure cannot judge whether the desilting was adequate. Fix it in the draft yourself.

Step 6 — send, and record what you sent. Send only to the training addresses, never a real colleague. Record the send time and any correction you made in Approval_Log, and leave Approval_Status blank: you have requested approval, not received it.

Step 7 — summarise any reply. If a reply arrives, use the summary prompt and record the manager's decision in their own wording. A polite acknowledgement is not an approval.

Independent checks: the WO-022 email shows 74 mm, the desilting and 28 mm in that order; neither email claims approval or closure; Approval_Log records what was sent and leaves approval blank.

Boundary: Copilot drafts the email; you send it. Once sent you cannot recall what the recipient has read. Never send training material to a real mailbox.

Lab 2.4 — weekly update to the supervisor (30 minutes)

Open your register and this lab's Starter.xlsx. Your two work orders were part of a larger week: six drain sections inspected against the same 50 mm maximum, of which three failed on the first visit. The supervisor wants one thing on a Friday afternoon — what happened, and what still needs them.

Step 1 — bring the week's data into your register. Copy Weekly_Inspections and Reinspections across with Move or Copy. Confirm your own WO-021 and WO-022 rows match what you recorded earlier: 32 mm pass, and 74 mm fail then 28 mm on reinspection.

Step 2 — work out the figures by hand first. Six inspections, three passed first visit, so the first-visit pass rate is 3 / 6 = 50.0%. Two failures were desilted and passed; one, WO-024 at DR-38 on 88 mm, is still open. If you do not know the right answers, you cannot tell whether the deck is right.

Step 3 — summarise the week in Excel. Select WeeklyInspections, open Copilot and run the summary prompt. It must treat 50 mm as a maximum, not a minimum, and must not average silt depths across different drain sections.

Step 4 — build the deck with Copilot in PowerPoint. Attach your register and run the five-slide prompt: title, the week at a glance, a bar chart of silt depth against the 50 mm threshold, the three failures with what was done, and what needs the supervisor.

Step 5 — add the workflow diagram. Insert a process diagram of the shared path and the two branches, and mark where this week's six orders sit: three closed, two awaiting approval, one awaiting rectification. A supervisor reading a list has to reconstruct the process mentally; the diagram does that for them.

Step 6 — check every figure against the source. Run the audit prompt, then check the traps yourself: no averaged silt depths, the pass rate described as what it counts, no desilted drain called closed while it awaits approval, and WO-024 present and visibly open.

Independent checks: all six work orders appear with the correct silt depth and status; the pass rate is 50.0%; the chart shows the 50 mm maximum with failing bars above it; WO-024 is shown as still open.

Boundary: every figure must come from the attached tables. The deck invents no cause, cost, deadline, photograph or approval, and calls no work order closed unless the table says so.

Module 3 — FMEA → DMAIC → SPC



Learning outcome: explain each method in plain language, select the correct method for the engineering question, use Copilot in Excel to produce a small output, and independently verify one result.

Sequence	Question	Primary output
1 · FMEA	What could fail and what needs attention?	Prioritised risk review and actions
2 · DMAIC	How do we investigate and improve performance?	Evidence trail, pilot and control plan
3 · SPC	Is process behaviour stable over time?	Control chart and investigation signals

FMEA — purpose and use

Failure Mode and Effects Analysis is a structured, proactive review of potential failures. Engineers use it during design, commissioning, process change, operation and periodic review. It helps a multidisciplinary team make risk reasoning visible before a failure causes harm or disruption.

FMEA field	Question	CW-401 example
Function	What must the item do?	Deliver liquid
Failure mode	How could the function fail?	Low delivery
Effect	What happens if it fails?	Incorrect dose
Possible cause	Why might it happen?	Restricted strainer
Current control	What prevents or detects it?	Delivery test

FMEA ratings and RPN

Rating	Meaning	Review caution
Severity (S)	Seriousness of the effect	Use the agreed consequence scale.
Occurrence (O)	Likelihood/frequency of the cause	Use evidence and a defined scale.
Detection (D)	Difficulty detecting before the effect	A higher score means harder to detect.

RPN = Severity × Occurrence × Detection. For FM01, 7 × 5 × 4 = 140. RPN helps rank discussion. It is not a probability, legal acceptance criterion or safety approval. Apply severity gates separately: FM04 has RPN 90 but Severity 9 and therefore still requires review.

Acting on FMEA results

- Confirm system boundary, failure logic and rating evidence.
- Rank RPNs and apply severity or regulatory gates.
- Assign an action, owner, due date and required evidence.
- Implement and verify the action.
- Reassess residual S/O/D only after evidence exists.

Lab 3.1 — FMEA document and RPN table (30 minutes)

Open courseware/Labs/Lab-3.1-FMEA-Document/Starter.xlsx. Four failure modes for the CW-401 dosing skid are already rated for Severity, Occurrence and Detection. The RPN and Severity flag columns are deliberately empty. Read the Rating_Guide sheet before you start, and note that Detection is inverted: a high D means detection is poor.

Step 1 — select the named table. Click any cell inside the FMEA table and confirm the Name Box shows FMEAInput. Copilot in Excel works far more reliably against a named table than a loose range, because the table tells it exactly where the data stops.

Step 2 — open Copilot and paste the prompt. On the Home ribbon click Copilot. Paste the whole prompt from Lab-3.1-Prompt.pdf. The prompt names the table explicitly and forbids outside knowledge; that single instruction is what stops Copilot inventing a fifth failure mode.

Step 3 — check the RPN arithmetic by hand before accepting it. FM-01 is 7 × 5 × 4 = 140. FM-02 is 8 × 4 × 6 = 192, the highest of the four. If Copilot shows a different number, or shows no arithmetic at all, reject the answer and ask it to show the multiplication per row.

Step 4 — insert the RPN chart. Ask for a clustered column chart of RPN by failure mode ID, ordered highest to lowest, with a title a reviewer would understand.

Step 5 — apply the severity gate yourself. Filter on the S column. FM-04 carries Severity 9 and the lowest RPN of the four at 54, and it still escalates. Write ESCALATE in its Severity flag cell. If Copilot did not flag it, that is the single most important correction you will make in this lab.

Step 6 — request actions and evidence gaps. Ask for one evidence request and one practical follow-up per row. Reject anything that reads like an approval: "approve the design" is wrong, "request the calibration certificate for the flow transmitter" is right.

Independent checks: FM-01 RPN = 140; FM-02 RPN = 192; FM-04 escalates on Severity 9 despite the lowest RPN; no fifth failure mode appeared; every supplied rating is unchanged. Only then open Worked-Reference.xlsx and compare.

Boundary: this is a drafting aid, not an approval. Never change a supplied rating, never state the skid is safe or fit for service, and never lower a rating because an action is planned. Only verified evidence of control effectiveness justifies a lower rating.

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Act as an FMEA facilitator. Use only FMEAInput. Preserve all rows and supplied ratings. Calculate RPN = Severity × Occurrence × Detection, rank the rows, flag Severity 9 or higher, and add one evidence request and action per row. Show formulas. Do not invent
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incidents or approve the design.

DMAIC — structured process improvement

Six Sigma is a data-based approach for reducing defects and variation. DMAIC is its five-phase roadmap for improving an existing process. It creates a disciplined link from problem definition to sustained control; filling the headings alone does not prove improvement.

Phase	Question	Output
Define	What problem matters, to whom and within what boundary?	Problem statement, scope and critical-to-quality measure
Measure	How does the process perform now?	Operational definition, data-quality check and baseline
Analyse	What patterns and possible causes should be tested?	Prioritised hypotheses and evidence gaps
Improve	Which supported change will be piloted?	Controlled change, owner and success criterion
Control	How will the gain be monitored and protected?	Metric, frequency, trigger, owner and response

Everyday Analyse tools

Tool	Use	Do not conclude
Pareto	Order problem categories from largest to smallest to focus effort.	The largest category is the root cause.
Fishbone / Ishikawa	Organise possible causes under Machine, Method, Measurement, People, Material and Environment.	Every branch is true.
5-Whys	Develop one causal chain and identify the next evidence check.	The fifth answer is automatically the root cause.

Worked rate comparison

Baseline = 36 deviations ÷ 400 opportunities = 9.0%. Pilot = 7 ÷ 200 = 3.5%. The observed change is −5.5 percentage points. Relative reduction is 5.5 ÷ 9.0 = 61.1%. The two measures are different. A short pilot after concurrent changes does not identify cause or prove sustained improvement.

Lab 3.2 — DMAIC improvement report (30 minutes)

Open courseware/Labs/Lab-3.2-DMAIC-Report/Starter.xlsx. Six weeks of dosing deviation data are supplied: three Baseline weeks and three Pilot weeks, each with Opportunities and Deviations. The rate column and the phase summary are empty.

Step 1 — understand the metric before calculating it. Deviation rate is deviations divided by opportunities, a proportion rather than a count. This matters because the pilot weeks carry roughly half the opportunities of the baseline weeks; comparing raw counts would flatter the result and would be wrong.

Step 2 — select WeeklyData and paste the prompt from Lab-3. -Prompt.pdf. This prompt carries an E-MAP block: it fixes the units, states the assumption that the deviation definition never changed across the six weeks, and limits the scope to this period only.

Step 3 — verify both phase rates by hand. Baseline is 36 ÷ 400 = 9.00 per cent. Pilot is 7 ÷ 200 = 3.50 per cent. The absolute change is 5.50 percentage points. If Copilot reports "a 61 per cent reduction" it has given you a relative change, which misleads in a report; ask for percentage points.

Step 4 — build the Pareto. On the Pareto sheet the five categories sum to 36, reconciling with the baseline deviations. Ask Copilot to add a cumulative percentage column and insert the chart. Confirm the top two categories carry most of the total, because that is what tells the team where to look first.

Step 5 — draft the five DMAIC rows: Define, Measure, Analyse, Improve, Control. Read the Analyse row hardest. Every causal statement must be labelled a hypothesis unless the data in front of you supports it.

Step 6 — force the limitation into the output. Ask directly what this data does not prove. The answer must state that a three-week pilot cannot demonstrate a sustained improvement, and that two concurrent changes cannot be separately attributed. Paste that into the Control row; a report without it is not review-ready.

Independent checks: 36 ÷ 400 = 9.00%; 7 ÷ 200 = 3.50%; change stated as 5.50 percentage points; Pareto counts sum to 36; the limitation appears in the output.

Boundary: do not invent a root cause, a cost saving or a completion percentage, and do not present a relative improvement as the absolute change. The status is draft; a competent reviewer decides whether the pilot is adopted.

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Act as a Six Sigma project assistant. Use only WeeklyData. Calculate Baseline and Pilot deviation rates. Create a five-row Define, Measure, Analyse, Improve, Control summary. Separate facts from hypotheses, show formulas, and state that the short pilot does not prove cause.
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Pareto worked example

In the optional reference data, Low delivery (55) and Test validity (16) contribute 71 of 93 baseline deviations, about 76%. This helps select a starting point; it does not prove why either category occurs.

Fishbone and 5-Whys worked example

Possible fishbone branches include Machine: pump wear/valve position; Method: setup/sequence; Measurement: calibration/timing; People: training/handover; Material: fluid condition/contamination; Environment: temperature/supply. A 5-Whys chain may move from low delivery → reduced outlet flow → possible restriction → possible filter loading. Stop and request inspection or differential-pressure evidence before treating the chain as fact.

SPC — Statistical Process Control

SPC uses measurements in time order to judge whether a process behaves consistently. The control chart distinguishes ordinary common-cause variation from special-cause signals that merit investigation. A signal identifies when to investigate; it does not diagnose the cause.

Chart element	Meaning
Time-ordered points	Measurements plotted in collection order
Centre line	Process average for the selected baseline
UCL/LCL	Expected process range calculated from baseline variation
Signal rule	Agreed pattern that triggers investigation

Common cause versus special cause

Variation	Plain meaning	Engineering response
Common cause	Small routine fluctuation produced by the current process system	Improve the system if performance is inadequate
Special cause	Unusual point or pattern indicating something changed	Check measurement, equipment, conditions and events

Control limits versus specification limits

Control limits come from process data and describe behaviour. Specification limits come from design, customer or regulatory requirements and describe acceptability. A 59.5 L/h result can satisfy the fictional 58.9–65.1 L/h specification while still being below the 61.4 L/h lower control limit and unusual for the process.

Process capability — Cp and Cpk in plain terms

Cp asks whether the estimated stable process spread could fit between the specification limits if centred: Cp = (USL − LSL) ÷ 6σ. Cpk also accounts for where the process mean actually sits: Cpk = min[(USL − mean), (mean − LSL)] ÷ 3σ. A value above 1 means the estimated six-sigma spread fits inside the limits; larger values provide more margin. Interpret only after confirming process stability, measurement quality, distribution and data sufficiency.

Lab 3.3 — SPC control chart and capability (30 minutes)

Open courseware/Labs/Lab-3.3-SPC-Control-Chart/Starter.xlsx. Sixteen subgroups of four delivery measurements are supplied. Subgroups 1 to 12 are Baseline and 13 to 16 are Monitoring. The Xbar and R columns are empty.

Step 1 — calculate one subgroup by hand first, so you know what right looks like. Subgroup 1: Xbar = (61.8 + 62.4 + 61.5 + 62.1) ÷ 4 = 61.950 and R = 62.4 − 61.5 = 0.900. This takes thirty seconds and it is the reason you will catch an error later.

Step 2 — select SPCData and paste the prompt from Lab-3. -Prompt.pdf. The prompt supplies the constants for n = 4: A2 = 0.729, D4 = 2.282 and d2 = 2.059. Copilot should not be guessing these values.

Step 3 — confirm the limits came from the baseline only. This is the step most people get wrong. Limits must be calculated on subgroups 1 to 12 and then frozen. If Copilot used all sixteen, the drifting monitoring points drag the limits down with them and the signal disappears. Expect Xbarbar ≈ 62.027 and Rbar ≈ 0.658, giving UCL = 62.027 + 0.729 × 0.658 = 62.507 and LCL = 62.027 − 0.729 × 0.658 = 61.547.

Step 4 — plot all sixteen subgroups against those frozen limits, with the centre line, UCL and LCL drawn as reference lines. The baseline points should sit inside the limits without a pattern; the monitoring points should fall away below the LCL.

Step 5 — read the signal correctly. Subgroup 13: Xbar = (61.5 + 61.2 + 61.4 + 61.0) ÷ 4 = 61.275, which is below the LCL of 61.547. All four monitoring subgroups are below the LCL and they trend steadily downwards rather than jumping once. A progressive drift like this suggests something like wear or fouling, but write the conclusion as a signal to investigate, not as a cause.

Step 6 — calculate capability and read it with caution. Expect sigma ≈ 0.320, Cp ≈ 3.13 and Cpk ≈ 1.01. Cp is roughly three times Cpk; that gap does not mean the process cannot hold the tolerance, it means the process is running off centre. State the larger caution too: capability is only meaningful while the process is stable, and these monitoring subgroups show it is not, so reporting Cpk alone here would mislead.

Independent checks: subgroup 1 Xbar = 61.950 and R = 0.900; limits calculated on subgroups 1-12 only; subgroup 13 Xbar = 61.275 below the LCL; Cp much greater than Cpk explained as off centre.

Boundary: a point outside a control limit is a signal to investigate, never a diagnosis and never proof that equipment is unsafe. Do not recalculate limits using monitoring data, and do not report capability as though the process were stable.

Act as an engineering data assistant. Use only PumpReadings. From the eight Baseline rows, calculate the mean, sample standard deviation, UCL and LCL. Flag T09–T12, create a line chart with fixed limits, show formulas, and give a three-sentence interpretation. Do not claim a root cause or safety conclusion.

Module 4 — Copilot Agents and Microsoft 365 Reasoning Agents



Learning outcome: configure and test a reusable multi-step FMEA-to-report workflow, then use Researcher and Analyst with citation and calculation checks.

When an agent is appropriate

Build an agent when the same bounded task repeats, required inputs and output are stable, acceptance tests exist, and the audience and knowledge permissions are known. Use an ordinary Copilot prompt for one-off or still-ambiguous work.

Agent component	Course configuration
Instructions	Collect → validate → calculate → retrieve → human gate → report
Knowledge	FMEA template, rating guide and test cases
Stop rules	Ask when S/O/D or source revision is missing; show conflicts
Output	Source fields, visible RPN, severity gate, evidence gap, action and audit note
Deployment	Preview, publish and share to the approved training audience

Guided FMEA Agent build

- Open Microsoft 365 Copilot and select Agents. Choose New agent or open the Copilot Studio agent builder available in the tenant.
- Name the agent “IES Pump FMEA Assistant” and state its narrow purpose.
- Paste the supplied FMEA-AGENT-INSTRUCTIONS.md content.
- Attach the supplied FMEA report template/rating reference if the environment allows knowledge files.
- Test FM01 with S7/O5/D4. Confirm RPN 140, evidence gap and human gate.

- Test a row with Detection missing. The agent must stop and request the value and scoring basis.
- Refine one instruction, rerun both tests, then publish/share only to the approved audience.

Reasoning agents — Researcher and Analyst

Agent	Use	Independent check
Researcher	Multi-source standards/technology comparison with citations	Open one citation and compare the passage with the claim
Analyst	File-based calculations, charts and data interpretation	Reproduce important formulas and denominators

Researcher prompt

Act as an engineering research assistant. Compare the supplied recording requirements with current approved guidance. Separate supplied facts from external findings. Cite each claim and list unresolved differences for engineering review.

Analyst prompt

Use only the six rows in WeeklyData. Check row count, phases and missing values. Calculate Baseline and Pilot rates, percentage-point change and relative reduction. Create a comparison chart, show formulas, state one limitation and one next evidence check. Do not claim cause or sustained improvement.

Independent checks: 9.0%; 3.5%; −5.5 percentage points; 61.1% relative reduction.

Module 5 — SharePoint Integration with Copilot



Learning outcome: organise a private engineering knowledge folder, ground an agent on that folder, test supported and missing-evidence questions, and pass only checked excerpts to another agent.

Prepared environment

Item	Value
Site	https://tertiaryinfotech.sharepoint.com/sites/IES-GenAI-for-Engineers
Library / folder	Documents / Engineering Knowledge Base
Sources	TR-401-Training-Extract.pdf; FMEA-Starter.xlsx; DMAIC-Starter.xlsx; SPC-Starter.xlsx
Agent	IES Engineering Knowledge Assistant

Create or inspect the knowledge base

- Confirm the site is private and inspect owners, members and visitors.
- Open Documents / Engineering Knowledge Base.
- Confirm exactly four synthetic sources and their revisions.
- Create or inspect the folder-scoped agent.
- Use instructions that require citations, separate facts from interpretation, and preserve missing evidence.

Agent test cases

Test	Expected result
Which fields must TR-401 record?	Test ID, timestamp, volume and duration, with citation to clause 1.
What approved delivery limit does TR-401 state?	Not specified; request the approved design requirement.

Compare the three workbook purposes.	FMEA risk review, DMAIC improvement summary, SPC monitoring; citation to each file.

End-to-end integration

Ask the SharePoint agent for a requirement. Open and verify the cited passage. A named engineer approves the excerpt as context. Pass only that checked excerpt to the FMEA or Report Writer agent. The generated document shows the source and reviewer. This preserves the human gate between retrieval and document generation.

Module 6 — 7-Step AI Security Framework



Learning outcome: apply every control to the Module 5 grounded answer and make an explicit approve, revise or reject decision.

Step	Action	Evidence
1 · Data classification	Classify input/output using organisational policy	Classification and prohibited fields
2 · Governance & access	Check source, user, agent scope and audience	Site/folder permissions and revision
3 · Secure prompting	Minimise sensitive data; resist embedded instructions	Final prompt and redactions
4 · Output verification	Open citation; reproduce calculation; check units	Pass/correction/unknown
5 · Human sign-off	Name competent reviewer and decision	Reviewer, date and conditions
6 · Traceability & audit	Retain prompt, sources, output and corrections	Reproducible record
7 · Awareness & incident response	Detect, stop, contain, report and improve	Incident path and retest

Applied review

- Use the SharePoint-agent workbook comparison and TR-401 missing-limit answer.
- Complete all seven rows in AI-SECURITY-EVIDENCE-LOG.csv.
- Open one citation and reproduce one FMEA, DMAIC or SPC calculation.
- Identify the embedded prompt-injection attempt as untrusted document content.
- Make one release decision: approve for named audience, revise and recheck, or reject and escalate.

Course close — personal AI Engineering Toolkit

Keep	Contents
Prompt patterns	RISE, E-MAP, constraints, personas and acceptance checks
Method quick references	FMEA → DMAIC → SPC
Agent assets	Instructions, knowledge and test cases
Grounding checks	Folder scope, citations and missing-evidence test
Security gate	Seven-step evidence log and release decision

One next action

Choose one small repeatable workflow. Name its evidence, required output, acceptance test and approving owner. Trial it with synthetic or approved data before introducing live confidential information.

Microsoft 365 learning links

- Microsoft 365 Copilot: https://m365.cloud.microsoft/chat
- Copilot in Excel: https://support.microsoft.com/en-us/office/get-started-with-copilot-in-excel
- Copilot in Word: https://support.microsoft.com/en-us/office/welcome-to-copilot-in-word
- Copilot in PowerPoint: https://support.microsoft.com/en-us/office/welcome-to-copilot-in-powerpoint
- Microsoft 365 Copilot agents: https://support.microsoft.com/en-us/topic/agents-in-microsoft-365-copilot
- Researcher: https://learn.microsoft.com/en-us/microsoft-365/copilot/researcher-agent
- Analyst: https://support.microsoft.com/en-US/Microsoft-365-Copilot/get-started-with-analyst-in-microsoft-365-copilot
- SharePoint agents: https://support.microsoft.com/en-us/office/get-started-with-agents-in-sharepoint

File map

Resource	Path
Slides	courseware/Slides/IES-GenAI-for-Engineers.pptx
All 16 Lesson Plan activities	courseware/Labs
FMEA workbook	courseware/Labs/Lab-3.1-FMEA-Document
DMAIC + SPC workbooks	courseware/Labs/Lab-3.2-DMAIC-with-SPC
Agent/SharePoint/security files	courseware/Trainer-Demonstrations
Optional quick references	PPT slides 163–186